

Ultrametric Quantum Kernels: Exact p -adic Feature Maps for Hierarchical Data

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Abstract

A quantum kernel compares data through the fidelity of an encoding, so the geometry the model can express is fixed before any training happens. Hierarchical data (taxonomies, ontologies, phylogenies) is not Euclidean but ultrametric, and we ask which quantum feature maps induce an exactly ultrametric kernel. We give two obstructions and a matching construction. First, once the tree has depth at least two, no product feature map induces a *strictly monotone* one, whatever its local dimensions: multiplicativity across digits limits any such encoding to a bounded resolution depth. Angle encoding is a corollary; computational-basis encoding is excluded separately, and is instructive because it *is* ultrametric while resolving nothing. Second, and independently, the Gram matrix of any exact realisation forces a dimension of at least $L^2 / \sum_{x,y} K(x,y)$ on L points; on the p -adic tree with a geometrically decaying profile of parameter $s > 1$ this exceeds 2^n for every $p \geq 3$ and all sufficiently deep trees, so no n -qubit pure-state encoding suffices there, including the entangling IQP and ZZ maps the first argument cannot reach. Third, *path states*, superpositions over a leaf's root-to-leaf ancestor chain with amplitudes read off the target profile, realise precisely the strictly monotone ultrametric kernels, on arbitrary rooted trees, and on $\mathbb{Z}/p^n$ they do so within an additive constant of that lower bound. We further show that global depolarising noise cannot break the strong triangle inequality at any rate, since it acts on distances as a non-decreasing affine map, and we measure the shot budget that finite sampling demands. Experiments on WordNet, Gene Ontology and the NCBI taxonomy confirm exactness to machine precision and separate ultrametricity from usable resolution: a basis encoding is ultrametric while resolving nothing. We claim no quantum speedup in kernel evaluation, and say so throughout; the contribution is a constraint on encoding design and an encoding, of optimal width on $\mathbb{Z}/p^n$, for models where no closed form survives.

Keywords

quantum machine learning, quantum kernels, ultrametric spaces, p -adic numbers, hierarchical representation learning

1. Introduction

A quantum kernel method embeds classical data through a feature map Φ and compares two points by the fidelity of their images,

$$K(x, y) = |\langle \Phi(x) | \Phi(y) \rangle|^2. \quad (1)$$

Everything a downstream model can express about the similarity of x and y passes through this one number, and the number is fixed by the encoding alone: whatever geometry Φ imposes is imposed before any parameter is trained [1, 2]. The choice of encoding is therefore not a preprocessing detail. It is the model's inductive bias.

Much of the data to which quantum machine learning is applied is hierarchical. Taxonomies, ontologies, phylogenies and type lattices are trees, and the natural distance on a tree is not Euclidean but *ultrametric*: distances obey the strong triangle inequality

$$d(x, z) \leq \max(d(x, y), d(y, z)), \quad (2)$$

which is strictly stronger than the ordinary triangle inequality. Equivalently, the points sit at the leaves of a tree and distance decreases with the depth of the lowest common ancestor.

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The canonical ultrametric space is the ring of p -adic integers $\mathbb{Z}_p$, whose valuation $v_p(x - y)$ counts matching least-significant base- p digits and so measures exactly that depth. This paper asks which quantum feature maps induce an exactly ultrametric kernel, and answers with two obstructions and a matching construction.

Theorem A: product encodings cannot. A map that factorises across digits induces a multiplicative kernel, and multiplicativity bounds how deep it can resolve the tree, whatever the local dimension: a product map is left with three kernel values, too few for a strictly monotone profile once $n \geq 2$. The argument extends to disjoint blocks of digits, where the bound is set by one block’s size, so it constrains a class of encodings rather than a list. Angle encoding is a corollary; computational-basis encoding is excluded by a separate one-line argument, and is the instructive case that is ultrametric while resolving nothing.

Theorem B: narrow encodings cannot either, whatever their structure. The Gram matrix of any exact realisation is positive semi-definite with unit diagonal, which forces its rank to at least $L^2 / \sum_{x,y} K(x,y)$ on L points. On the regular p -ary tree with a geometric profile of parameter $s > 1$ this exceeds 2^n for every $p \geq 3$ once n passes an explicit threshold: beyond it, no n -qubit pure-state encoding realises that kernel on $\mathbb{Z}/p^n$, entangling or not. This is what rules out the IQP and ZZ feature maps Theorem A cannot reach.

Theorem C: path states can, at essentially the optimal width on $\mathbb{Z}/p^n$. Writing $\text{anc}_i(x)$ for the depth- i ancestor of a leaf x , the state $\Phi_f(x) = \sum_i a_i |\text{anc}_i(x)\rangle$ realises *every* strictly monotone ultrametric kernel exactly, with amplitudes read off the target profile. It needs $\lceil \log_2 |V| \rceil$ qubits on a tree of $|V|$ nodes, $n + 1$ non-zero amplitudes and $O(n \log |V|)$ gates. On $\mathbb{Z}/p^n$ with $s > 1$ that matches Theorem B to within an additive constant; on a general tree the two quantities are not directly comparable.

We also record a fourth fact with a two-line proof: global depolarising noise acts on the induced distance as a non-decreasing affine map, so it cannot break the strong triangle inequality or level constancy at any error rate (Proposition 7.1). What it destroys is contrast rather than geometry. Finite sampling behaves in the opposite way and breaks the inequality immediately, for a reason specific to ultrametric spaces that Section 8 makes precise.

What this paper does not claim. The kernel of Theorem 6.1 is $K(x,y) = f(\lambda(x,y))$, and a lowest common ancestor is computable classically in time linear in the depth of the tree. **There is no quantum speedup in kernel evaluation here, and we claim none.** We say so in the introduction rather than leaving a reader to discover it, because it determines how the rest should be read. The contribution is a no-go that constrains current encoding practice; a dimension bound that applies to arbitrary encodings and so survives future designs; and an explicit encoding, of optimal width on $\mathbb{Z}/p^n$, usable as the input stage of deeper quantum models, where no closed form survives the subsequent layers.

Contributions. **C1**, a no-go for product feature maps (Theorem 4.1) bounding the resolution depth of any block-product map by one block’s size and collapsing to three kernel values for genuine product maps. **C2**, a dimension lower bound for arbitrary feature maps (Theorem 5.1), ruling out every n -qubit encoding of the geometric profile on $\mathbb{Z}/p^n$ for $p \geq 3$ past an explicit threshold. **C3**, an exact construction (Theorem 6.1) realising precisely the strictly monotone ultrametric kernels, matching C2 on $\mathbb{Z}/p^n$ to within an additive constant, with a converse characterising every realisation with non-negative overlaps up to isometry, read modulo the populated levels. **C4**, its generalisation to arbitrary rooted trees, with $\mathbb{Z}/p^n$ as the specialisation in which the profile ties to $|\cdot|_p$. **C5**, an empirical study on WordNet, Gene Ontology and the NCBI

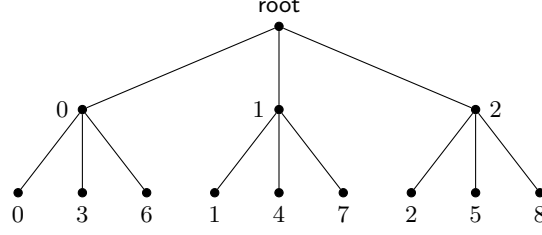


Figure 1: The regular 3-ary tree of depth 2 for Example 2.1. Internal nodes are labelled by the residue $x \bmod 3$ that defines them; leaves by $x \in \mathbb{Z}/9$. Two leaves meet at depth $v(x, y) = v_p(x - y)$: the leaves 2 and 5 share a parent, the leaves 2 and 4 share only the root.

taxonomy separating ultrametricity from usable resolution, mapping the expressivity/hierarchy trade-off of a re-uploading extension, and quantifying the shot budget required.

Availability. Every theorem has a corresponding numerical check, cited by test name at the point of use, and every experimental number in the text is generated from the experiment outputs rather than transcribed. The library, the five experiment scripts, the verification suite and the sources of this paper are at <https://github.com/pgiacome/aiqxqia2026>.

2. Preliminaries

2.1. p -adic integers and the regular tree

Fix a prime p and an integer $n \geq 1$. Write $X = \mathbb{Z}/p^n$ and expand each $x \in X$ in base p as $x = \sum_{i=1}^n x_i p^{i-1}$ with digits $x_i \in \{0, \dots, p-1\}$. The p -adic valuation of a non-zero integer is the exponent of the largest power of p dividing it, and the p -adic absolute value is $|m|_p = p^{-v_p(m)}$. On X we use the truncated valuation

$$v(x, y) = \min(v_p(x - y), n) = \#\{\text{matching least-significant base-}p \text{ digits of } x \text{ and } y\}. \quad (3)$$

Identify X with the leaves of the regular p -ary tree of depth n , a leaf x sitting under the chain of residue classes $x \bmod p$, $x \bmod p^2, \dots$. Then the depth- i ancestor of x is the class $x \bmod p^i$, and $v(x, y)$ is exactly the depth of the lowest common ancestor of x and y .

Example 2.1. Take $p = 3$, $n = 2$, so $X = \{0, \dots, 8\}$. The leaves $x = 2$ and $y = 5$ have $x - y = -3$, hence $v_p(x - y) = 1$: they agree in the least-significant digit ($2 \equiv 5 \bmod 3$) and differ in the next, so they meet at depth 1. The leaves 2 and 4 differ already in the least-significant digit and meet only at the root, at depth 0. Figure 1 draws the tree.

2.2. Ultrametric spaces

A metric d on a set X is an *ultrametric* if it satisfies Eq. (2) for all x, y, z . Two consequences matter here. First, every triangle is isocles with its two longest sides equal, so a constant fraction of triples meets Eq. (2) *with equality*. That fact turns out to govern how ultrametricity behaves under sampling noise (Section 8). Second, the closed balls of an ultrametric space are nested or disjoint, never partially overlapping, which is precisely the combinatorics of a rooted tree.

Real hierarchical data is rarely exactly ultrametric, and quantifying the discrepancy is an active question [3]. Our concern is the complementary one: given that the target geometry *is* ultrametric, which encodings can represent it without distortion.

2.3. Quantum feature maps and fidelity kernels

A quantum feature map is a function $\Phi : X \rightarrow \mathbb{C}^D$ with $\|\Phi(x)\| = 1$, realised in practice by a state-preparation circuit on $\lceil \log_2 D \rceil$ qubits. Its fidelity kernel is Eq. (1). Writing $G_{xy} = \langle \Phi(x) | \Phi(y) \rangle$ for the Gram matrix of the feature vectors, we have $K = |G|^{\circ 2}$ entrywise, and G is positive semi-definite with $G_{xx} = 1$. Since the Schur product of positive semi-definite matrices is positive semi-definite, K is a valid kernel for any Φ .

Two families of encoding recur below. An *angle encoding* assigns one qubit per digit, $|\psi_i(a)\rangle = \cos(\theta_a/2)|0\rangle + \sin(\theta_a/2)|1\rangle$, and takes the tensor product. A *ZZ (or IQP-style) feature map* applies a data-dependent diagonal phase between layers of Hadamards, $\Phi(x) = (U_{\phi(x)} H^{\otimes n})^r |0\rangle^{\otimes n}$, with the phase carrying both single-qubit and pairwise terms [2]. The pairwise terms make the ZZ map entangling, which is exactly why Theorem 4.1 does not apply to it and Theorem 5.1 must.

3. Ultrametric Quantum Kernels

Throughout, T is a finite rooted tree whose leaves all sit at depth n (a leaf that is shallower is extended by a chain of unary nodes, which changes no lowest common ancestor). For a leaf x and $0 \leq i \leq n$ we write $\text{anc}_i(x)$ for its depth- i ancestor, so $\text{anc}_0(x)$ is the root and $\text{anc}_n(x) = x$, and $\lambda(x, y) = \max\{i : \text{anc}_i(x) = \text{anc}_i(y)\}$ for the depth at which two leaves meet. On the regular p -ary tree, $\lambda = v$ of Eq. (3).

Definition 3.1 (Ultrametric kernel). A kernel $K : X \times X \rightarrow [0, 1]$ is *ultrametric with profile* f if there is a function $f : \{0, \dots, n\} \rightarrow [0, 1]$ with $f(n) = 1$ such that $K(x, y) = f(\lambda(x, y))$ for all leaves x, y . The profile is *strictly monotone* if f is strictly increasing and $f(0) > 0$.

Being ultrametric with *some* profile is the same as being constant on each level of the lowest-common-ancestor relation, a property we call *level constancy*. Level constancy alone is not enough to make the induced distance an ultrametric: $f = (0.9, 0.1, 1)$ on a depth-2 tree is level-constant, yet two leaves meeting at depth 1 are further apart than either is from a leaf meeting them at the root. Monotonicity is what supplies the strong triangle inequality.

Lemma 3.2. *If K is ultrametric with a non-decreasing profile f satisfying $f(v) < 1$ for every $v < n$, then $d = 1 - K$ is an ultrametric on the leaves of T . Without the second hypothesis d is a pseudo-ultrametric: $f \equiv 1$ gives $d \equiv 0$.*

Proof. Lowest-common-ancestor depth satisfies $\lambda(x, z) \geq \min(\lambda(x, y), \lambda(y, z))$, because the ancestors of a leaf form a chain: if x and y agree down to depth i and y and z agree down to depth i , then x and z agree down to depth i . Applying the non-decreasing f preserves this and reverses it under $d = 1 - f$, giving Eq. (2). Non-degeneracy follows from $f(n) = 1$ and $f(v) < 1$ for $v < n$. $\square$

3.1. Two diagnostics, because one is not enough

It is tempting to certify an encoding by counting violations of Eq. (2). That test is necessary but far from sufficient, and the gap is not a technicality. Computational-basis encoding sends distinct leaves to orthogonal states, so its kernel is the identity and its induced distance is the discrete metric — which is an ultrametric, with zero violations, while collapsing the hierarchy to a single bit of information. It is the degenerate corner of Definition 3.1, with $f = \mathbf{1}[v = n]$.

We therefore report two quantities throughout.

Definition 3.3 (Level constancy and resolution depth). Call a level v *populated* if some pair of leaves has $\lambda(x, y) = v$, and let $V(K)$ collect them. The *level constancy* is

$$\max_{v \in V(K)} (\max K_v - \min K_v), \quad K_v = \{K(x, y) : \lambda(x, y) = v\},$$

which vanishes exactly when K is constant on each populated level; for a fidelity kernel, whose diagonal is 1, that is exactly Definition 3.1. Writing μ_v for the mean of K over level v , the *resolution depth* is the number of runs in the sequence $(\mu_v)_{v \in V(K)}$ after merging consecutive entries differing by at most τ ; we take $\tau = 10^{-6}$ throughout, a scale no realistic shot budget could resolve (Section 8.5). It equals $|V(K)|$ for a strictly monotone profile whose consecutive values differ by more than τ , and 2 for the discrete metric. Pairs with $x = y$ count, so $n \in V(K)$ always.

Restricting to populated levels matters on padded and irregular trees, where some v may carry no pair of leaves at all. At such a v the profile is not determined by K : two profiles agreeing on $V(K)$ induce the same kernel. Every statement below that recovers f from K is therefore to be read modulo $V(K)$.

An encoding is useful for hierarchical data only if it scores well on both: level constancy zero, resolution depth full. Theorem 6.1 exhibits a family that does, for every strictly monotone profile and every rooted tree. It is not the only one — Remark 6.2 constructs a second realisation of the same kernel that is not isometric to it, and Example 5.3 gives a one-qubit realisation on a depth-one tree — but among the encodings in common use it is the only one that scores well on both, which is what Section 8 measures.

3.2. Product and block-product feature maps

Definition 3.4 (Product feature map). A feature map on $\mathbb{Z}/p^n$ is a *product map* if $\Phi(x) = \bigotimes_{i=1}^n |\psi_i(x_i)\rangle$ for unit vectors $|\psi_i(a)\rangle \in \mathbb{C}^{d_i}$, where factor i depends only on digit i and the local dimensions d_i are arbitrary. Writing $g_i(a, b) = |\langle \psi_i(a) | \psi_i(b) \rangle|^2$, we have $g_i(a, a) = 1$ and

$$K(x, y) = \prod_{i=1}^n g_i(x_i, y_i). \quad (4)$$

Definition 3.5 (Block-product feature map). Let $\mathcal{B}$ be a partition of the levels $\{1, \dots, n\}$ into consecutive blocks. A *block-product map* is $\Phi(x) = \bigotimes_{B \in \mathcal{B}} |\psi_B(x_B)\rangle$, where x_B is the tuple of digits in block B . Its kernel factorises across blocks in the same way. A product map is the case in which every block is a single level.

Block-product maps let a bounded number of levels interact arbitrarily while keeping the tensor structure across blocks. They are the natural strengthening of Definition 3.4, and Theorem 4.1 handles both at once. They do not, however, cover maps such as the ZZ feature map, whose all-pairs couplings leave no tensor factorisation at all; that case is Section 5.

4. A No-Go Theorem for Product Encodings

Theorem 4.1 (Product no-go). *Let Φ be a block-product feature map on $\mathbb{Z}/p^n$ (Definition 3.5) whose fidelity kernel is ultrametric with profile f . Put $v^* = \min\{v : f(v) > 0\}$ and assume $v^* < n$, and let B^* be the block containing level $v^* + 1$. Then for every block B all of whose levels exceed $\max B^*$, and all digit tuples a, b ,*

$$|\langle \psi_B(a) | \psi_B(b) \rangle| = 1. \quad (5)$$

Consequently $f(v) = 1$ for every $v \geq \max B^$: the map resolves the tree only to depth $\max B^*$, and its resolution depth is at most $|B^*| + 2$.*

For a product map every block is a single level, so $\max B^ = v^* + 1$ and f takes at most the three values $\{0, f(v^*), 1\}$. In particular no product map realises a strictly monotone profile once $n \geq 2$.*

Proof. The argument uses exactly one property of the encoding: fidelity is multiplicative across tensor factors, Eq. (4).

Choose leaves x, y that agree on levels $1, \dots, v^*$, differ at level $v^* + 1$, and agree at every level above that; such a pair exists because $v^* < n$. Then $\lambda(x, y) = v^*$, and every factor except B^* contributes $g(x, x) = 1$, so

$$f(v^*) = K(x, y) = |\langle \psi_{B^*}(x_{B^*}) | \psi_{B^*}(y_{B^*}) \rangle|^2 > 0. \quad (6)$$

Now fix a block B whose levels all exceed $\max B^*$ and digit tuples $a \neq b$ for it, and re-choose x, y as above with the further constraint $x_B = y_B = a$, which is possible because B does not meet levels $1, \dots, v^* + 1$. Let y' agree with y everywhere except on B , where $y'_B = b$. Because x and y' still first differ at level $v^* + 1$, and B lies strictly above B^* , we still have $\lambda(x, y') = v^*$ and therefore $K(x, y') = f(v^*)$. Multiplicativity gives

$$K(x, y') = |\langle \psi_{B^*}(x_{B^*}) | \psi_{B^*}(y_{B^*}) \rangle|^2 \cdot |\langle \psi_B(a) | \psi_B(b) \rangle|^2 = f(v^*) \cdot |\langle \psi_B(a) | \psi_B(b) \rangle|^2.$$

Since $f(v^*) > 0$ by Eq. (6), this forces $|\langle \psi_B(a) | \psi_B(b) \rangle|^2 = 1$, which is Eq. (5). Equality in Cauchy–Schwarz for unit vectors then gives $|\psi_B(b)\rangle = e^{i\theta} |\psi_B(a)\rangle$: block B carries no information about the data.

Every block beyond B^* being trivial, K depends only on the digits in blocks up to and including B^* , that is, on levels $1, \dots, \max B^*$. Two leaves agreeing down to depth $\max B^*$ therefore agree in every non-trivial factor and have $K(x, y) = 1$, so $f(v) = 1$ for all $v \geq \max B^*$. Together with $f(v) = 0$ for $v < v^*$, only the levels $v^* \leq v < \max B^*$ remain unconstrained, and they number at most $|B^*|$; the resolution depth is therefore at most $|B^*| + 2$.

For a product map $|B^*| = 1$ and $\max B^* = v^* + 1$, so the profile is 0 below v^* , equal to $f(v^*)$ at v^* , and 1 from $v^* + 1$ on: three values. Now suppose the profile is strictly monotone. Then $f(0) > 0$, so $v^* = 0$ and the value 0 is never attained, leaving only $\{f(0), 1\}$ — two values. A strictly monotone profile takes $n + 1$ distinct values, so $n + 1 \leq 2$, that is $n \leq 1$. Hence no product map realises a strictly monotone profile once $n \geq 2$. $\square$

The proof turns on multiplicativity alone. It is therefore indifferent to the local dimensions d_i , to whether the local states are real or complex, and to how they were chosen, which is what makes it a statement about a class of encodings rather than about particular instances.

Remark 4.2 (The three-value claim is genuinely about product maps). Nothing in Definition 3.5 forbids the single-block partition $\mathcal{B} = \{\{1, \dots, n\}\}$, under which the path state of Theorem 6.1 is itself a block-product map realising $n + 1$ distinct values. The theorem is consistent with this, since $\max B^* = n$ makes its conclusion vacuous. What it constrains is how far a map can resolve *beyond* the block containing the first informative level, which is severe when blocks are small; only at $|B^*| = 1$ does it collapse to three values. Sharpening slightly: for a strictly monotone profile $v^* = 0$, so the zero run disappears and the cap is $|B^*| + 1$; requiring $n + 1 \leq |B^*| + 1$ then forces $|B^*| = n$, a single block. A block-product map realises a strictly monotone ultrametric kernel only by not factorising at all.

Corollary 4.3. *On $\mathbb{Z}/p^n$ with $n \geq 2$, neither angle encoding with one qubit per digit nor any first-order Pauli-Z feature map — whose phase is a sum of single-digit terms, so it is a product map — induces a strictly monotone ultrametric kernel.*

Computational-basis encoding is excluded too, but by a shorter route that sits outside Theorem 4.1’s hypothesis rather than inside it. Its kernel is the identity, so $f(0) = 0$ and the profile is not strictly monotone; equivalently $v^* = n$. It is the case that can be mistaken for a success: the induced distance is the discrete metric, which satisfies Eq. (2) with zero violations while resolving nothing. That is why Section 3.1 reports resolution depth alongside violations.

Remark 4.4 (What this theorem does not cover). A ZZ or IQP feature map applies all-pairs couplings between qubits, so its state does not factorise across digits and Eq. (4) fails. Theorem 4.1 says nothing about it. We do not patch the gap by weakening the claim; we close it in Section 5 with an argument that ignores the structure of the encoding entirely.

Numerical verification. In all twenty trials — five seeds at each of $(p, n) = (2, 3), (3, 2), (2, 4)$ and $(5, 2)$ — randomly drawn product maps failed level constancy (`test_theorem_a_random_product_maps_are_never_strictly_monotone`). A product map built to satisfy the conclusion, with one non-trivial level and the rest trivial, yields exactly two distinct kernel values as predicted (`test_theorem_a_ultrametric_product_map_resolves_only_one_level`), and switching on a second non-trivial level destroys level constancy, which is the contradiction the proof derives (`test_theorem_a_a_nontrivial_deep_factor_destroys_ultrametricity`).

5. A Dimension Lower Bound for Arbitrary Encodings

Theorem 4.1 exploits the structure of the encoding. The bound in this section uses none: it constrains any map whatsoever into a space of given dimension, and so applies to entangling encodings, hardware-efficient ansätze, and designs not yet proposed.

Theorem 5.1 (Dimension lower bound). *Let $|X| = L < \infty$ and let $\Phi : X \rightarrow \mathbb{C}^D$ send its points to unit vectors with $|\langle \Phi(x) | \Phi(y) \rangle|^2 = K(x, y)$. Then*

$$D \geq \text{rank } G \geq \frac{L^2}{\sum_{x,y} K(x, y)}, \quad (7)$$

where G is the Gram matrix of the $\Phi(x)$.

Proof. The vectors $\Phi(x)$ lie in $\mathbb{C}^D$, so $\text{rank } G \leq D$. Let $r = \text{rank } G$ and let $\mu_1, \dots, \mu_r$ be the non-zero eigenvalues of G , which are real and non-negative because G is positive semi-definite. Since $\|\Phi(x)\| = 1$ we have $G_{xx} = 1$ and hence $\text{tr } G = \sum_k \mu_k = L$. By Cauchy–Schwarz applied to the vector of eigenvalues,

$$\|G\|_F^2 = \sum_{k=1}^r \mu_k^2 \geq \frac{1}{r} \left(\sum_{k=1}^r \mu_k \right)^2 = \frac{L^2}{r}. \quad (8)$$

Finally $\|G\|_F^2 = \sum_{x,y} |G_{xy}|^2 = \sum_{x,y} K(x, y)$ by the definition of the fidelity kernel. Rearranging Eq. (8) gives Eq. (7). $\square$

The bound is a statement about how *concentrated* the kernel is. A kernel close to the all-ones matrix has $\sum_{x,y} K \approx L^2$ and demands only one dimension, correctly: a near-constant kernel is realisable by a single state. A kernel close to the identity has $\sum_{x,y} K \approx L$ and demands L dimensions, also correctly. Strictly monotone ultrametric profiles decay fast enough to sit near the second regime, and the following corollary makes that quantitative.

Corollary 5.2 (When no n -qubit encoding suffices). *On the regular p -ary tree of depth n with the geometric profile $f(v) = p^{-(n-v)s}$, put*

$$S(p, s, n) = 1 + \frac{p-1}{p} \sum_{m=1}^n p^{m(1-s)}. \quad (9)$$

Then any exact realisation of the corresponding kernel by unit vectors needs $D \geq p^n / S(p, s, n)$, that is, at least $n \log_2 p - \log_2 S(p, s, n)$ qubits. Consequently no n -qubit pure-state encoding realises it whenever

$$p^n > 2^n S(p, s, n). \quad (10)$$

Three regimes make Eq. (10) concrete. For $s > 1$ the sum converges and $S(p, s, n) \leq S(p, s) < \infty$ uniformly in n , so Eq. (10) holds for every $p \geq 3$ and every

$$n > n_0(p, s) = \frac{\log_2 S(p, s)}{\log_2 p - 1}. \quad (11)$$

For $s \leq 1$ the bound weakens but does not vanish: $S(p, s, n) = \Theta(n)$ at $s = 1$ and $\Theta(p^{n(1-s)})$ below it, so p^n/S still grows like p^{ns} and Eq. (10) holds for all large n whenever $s > \log_p 2$.

Proof. Fix a leaf x . The number of leaves y with $\lambda(x, y) = v$ is $p^{n-v} - p^{n-v-1}$ for $v < n$, and 1 for $v = n$. Hence $\sum_y f(\lambda(x, y)) = S(p, s, n)$ after substituting $m = n - v$, and by symmetry $\sum_{x,y} K = p^n S(p, s, n)$. Inserting this and $L = p^n$ into Eq. (7) gives the bound, and comparing it with 2^n gives Eq. (10). The geometric series $\sum_m p^{m(1-s)}$ converges for $s > 1$, is the constant sum n at $s = 1$, and is dominated by its last term $p^{n(1-s)}$ below, which gives the three regimes. $\square$

Example 5.3. At $p = 3$, $s = 2$ we get $S \rightarrow 4/3$, so $n_0 = \log_2(4/3)/(\log_2 3 - 1) \approx 0.71$ and the requirement $q \geq 1.585n - 0.415$ exceeds n for every $n \geq 1$. At $n = 5$ the bound is 7.51 qubits against the 5 an angle or ZZ encoding provides; at $p = 5$ it is 11.3. Figure 3(b) plots the crossing.

The threshold is not decorative. At $p = 3$, $s = 1.01$ we get $S \approx 61.3$ and $n_0 \approx 10.2$, so the corollary says nothing for small trees — and indeed the bound at $n = 1$ is 0.85 qubits, while three qubit states with pairwise fidelity $3^{-1.01} \approx 0.33$ exist on the Bloch sphere. A *single* qubit realises that strictly monotone ultrametric kernel exactly. The hypothesis $n > n_0(p, s)$ is therefore not cosmetic: some depth hypothesis is unavoidable.

Remark 5.4 (Scope). Corollary 5.2 is a statement about the geometric profile with parameter s , not about strictly monotone profiles in general. A profile close to constant has $\sum_{x,y} K \approx L^2$, so Eq. (7) degrades to $D \geq \Omega(1)$ and narrow encodings are not excluded. We do not know whether they exist in that regime.

Remark 5.5 (What $s > 1$ does and does not buy). It buys the n -independent threshold Eq. (11), not the exclusion itself. At $s = 1$ every term of Eq. (9) equals 1 and $S(p, 1, n) = 1 + n(p-1)/p$ grows with the depth, so no uniform threshold exists — but $p^n/\Theta(n)$ still overtakes 2^n , and the exclusion holds from $n = 3$ at $p = 3$ and already at $n = 1$ at $p = 5$. Collecting the branches, the exclusion holds for all large n exactly when

$$\min(s, 1) \log p > \log 2, \quad \text{that is} \quad \min(s, 1) > \log_p 2, \quad (12)$$

which reduces to $p \geq 3$ when $s \geq 1$ and to $s > \log_p 2$ below. It correctly excludes $p = 2$ at every s , in agreement with Remark 5.6 (test_theorem_b_exclusion_criterion).

Remark 5.6 (Theorems A and B are independent). At $p = 2$ the denominator of Eq. (11) vanishes: the bound reads $q \geq n - \log_2 S(2, s)$, which is below n for every s and every n ($\log_2 S(2, 2) = \log_2 1.5 \approx 0.585$). Theorem 5.1 therefore leaves n -qubit encodings on binary trees open, and it is Theorem 4.1's structural argument that closes the product case there. Conversely Theorem 4.1 says nothing about entangling maps at any radix. Neither theorem subsumes the other, and we present them as covering different regimes rather than as a chain (test_theorem_b_does_not_subsume_theorem_a_at_radix_two).

Numerical verification. The closed form Eq. (9) matches the empirical row sum to machine precision (test_theorem_b_closed_form_row_sum); the bound exceeds 2^n for $(p, n) \in \{(3, 2), (3, 3), (3, 4), (5, 2)\}$ (test_theorem_b_rules_out_every_n_qubit_encoding); and the ZZ feature map, which lives in 2^n dimensions, falls below the bound and indeed fails to reproduce the profile (test_theorem_b_zz_feature_map_is_below_the_bound).

Two cautions on applying Eq. (7) in practice. Evaluated on a subsample of L' leaves it remains valid, since restricting the point set can only make the kernel easier to realise, but it is capped

at L' and so may understate the full-tree bound, saturating at $\log_2 L'$ qubits. The figures in Section 8 therefore use the closed form. And the bound constrains dimension, not circuit depth: it says nothing about how hard the required state is to prepare, which is the subject of Section 6.

6. Exact Realisation by Path States

The two obstructions leave a gap: strictly monotone ultrametric kernels need many dimensions and cannot be factorised across digits. The construction in this section fills the gap exactly, and turns out to characterise the whole family.

Theorem 6.1 (Path states). *Let T be a finite rooted tree on node set V with all leaves at depth n . For every strictly increasing $f : \{0, \dots, n\} \rightarrow (0, 1]$ with $f(n) = 1$, set*

$$a_i^2 = \sqrt{f(i)} - \sqrt{f(i-1)} \quad (0 \leq i \leq n), \quad \sqrt{f(-1)} := 0, \quad (13)$$

and define the path state

$$\Phi_f(x) = \sum_{i=0}^n a_i |\text{anc}_i(x)\rangle \in \mathbb{C}^V. \quad (14)$$

Then $\Phi_f(x)$ is a unit vector and, for all leaves x, y ,

$$|\langle \Phi_f(x) | \Phi_f(y) \rangle|^2 = f(\lambda(x, y)) \quad \text{exactly.} \quad (15)$$

Conversely, every kernel that is ultrametric with a strictly monotone profile arises this way: among realisations whose pairwise overlaps are real and non-negative, Φ_f is unique up to a global isometry. Dropping that restriction, uniqueness fails even up to isometry and per-point phases (Remark 6.2).

Proof. Strict monotonicity makes each a_i^2 in Eq. (13) positive, so the a_i are well defined and real. The sum telescopes, $\sum_{i=0}^n a_i^2 = \sqrt{f(n)} = 1$, so $\Phi_f(x)$ is a unit vector.

For the overlap, the key structural fact is that two leaves share precisely their ancestors down to the depth at which they meet, and no deeper: $\text{anc}_i(x) = \text{anc}_i(y)$ if and only if $i \leq \lambda(x, y)$. The basis vectors $|u\rangle$ for $u \in V$ being orthonormal,

$$\langle \Phi_f(x) | \Phi_f(y) \rangle = \sum_{i=0}^n a_i^2 \mathbf{1}[\text{anc}_i(x) = \text{anc}_i(y)] = \sum_{i \leq \lambda(x, y)} a_i^2 = \sqrt{f(\lambda(x, y))}, \quad (16)$$

again by telescoping. Squaring the modulus gives Eq. (15).

Two remarks on Eq. (16). The cross terms vanish because $\text{anc}_i(x)$ and $\text{anc}_j(y)$ sit at different depths when $i \neq j$, so they are distinct nodes and hence orthogonal basis vectors. And $a_i^2 = \sqrt{f(i)} - \sqrt{f(i-1)}$ is forced by Eq. (16) read at successive levels, so f determines the amplitudes.

For the converse, let Ψ be a realisation whose Gram matrix G^Ψ has real non-negative entries. Then $G_{xy}^\Psi = \sqrt{f(\lambda(x, y))}$ exactly, which by Eq. (16) is the Gram matrix of Φ_f . Two families of vectors with equal Gram matrices are related by an isometry of their spans, which is the standard argument. No phase freedom remains inside this class: since $f > 0$ every overlap is strictly positive, and a non-constant per-point phase would move G^Ψ out of it. $\square$

Remark 6.2 (How far uniqueness actually goes). Two weakenings are needed, and the second is not cosmetic.

First, the converse cannot say “unique up to relabelling of V ”. Rotating every $\Phi_f(x)$ by a fixed unitary, or multiplying each by its own phase, leaves the fidelity kernel unchanged while destroying the reading of the basis as tree nodes (test_theorem_c_uniqueness_up_to_isometry_and_phase).

Table 1

Resources for the path-state encoding, on the full WordNet hierarchy rather than the depth-8 truncation Section 8 applies to the baselines. Qubits is for the dense $\mathbb{C}^V$ layout; the gate count is for the level-indexed layout of Section 6.1, which costs about $\log_2 n$ more qubits. The two columns therefore describe two circuits, not one.

	nodes $ V $	qubits	non-zero amplitudes	preparation gates
general tree	$ V $	$\lceil \log_2 V \rceil$	$n + 1$	$O(n \log V)$
$\mathbb{Z}/p^n$	$\frac{p^{n+1}-1}{p-1}$	$\approx n \log_2 p$	$n + 1$	$O(n^2 \log p)$
WordNet, padded	673 790	20	20	—
WordNet, collapsed	132 231	18	20	—

The node-labelled form Eq. (14) is the canonical representative of its isometry class, not the only member.

Second, and less obviously, the positivity hypothesis cannot be dropped. A fidelity kernel fixes only the *moduli* $|G_{xy}|$, and these do not determine G up to phases: the Bargmann invariants $G_{xy}G_{yz}G_{zx}$ are invariant under $\Psi(x) \mapsto e^{i\vartheta_x}U\Psi(x)$ yet are not functions of the moduli. Concretely, take $n = 1$, $p = 3$, $f(0) = t = 1/4$ and

$$G^\Psi = \begin{pmatrix} 1 & \sqrt{t} & i\sqrt{t} \\ \sqrt{t} & 1 & \sqrt{t} \\ -i\sqrt{t} & \sqrt{t} & 1 \end{pmatrix},$$

which is positive semi-definite with spectrum $\{0.134, 1, 1.866\}$ and realises the same kernel as Φ_f . Its Bargmann invariant is $-\frac{1}{8}i$ against $+\frac{1}{8}$ for the path state, so no isometry and no choice of phases carries one to the other. So $f \mapsto \Phi_f$ is a bijection onto the strictly monotone ultrametric kernels only after two restrictions: to realisations with non-negative overlaps, and — since profiles agreeing on the populated levels $V(K)$ induce the same kernel (Definition 3.3) — modulo $V(K)$. On a tree with an unpopulated level, injectivity genuinely fails: a root with two unary-padded children of depth 2 admits both $f = (\frac{1}{4}, \frac{1}{2}, 1)$ and $g = (\frac{1}{4}, \frac{3}{4}, 1)$ for the same kernel, because no pair of leaves meets at depth 1 (test_theorem_c_converse_needs_non_negative_overlaps, test_theorem_c_profile_is_determined_only_on_populated_levels).

Remark 6.3 (Positive semi-definiteness for free). Let A be the leaf-by-node incidence matrix with $A_{xu} = a_{\text{depth}(u)}$ if u is an ancestor of x and 0 otherwise. Then Eq. (16) says $\sqrt{K} = AA^\top$ entrywise, so $K = (AA^\top)^{\circ 2}$ is a Schur square of a positive semi-definite matrix and hence positive semi-definite. This re-derives, in one line, the classical fact that monotone ultrametric kernels are positive semi-definite.

6.1. Resources

The path state lives in $\mathbb{C}^V$, so a dense layout needs $\lceil \log_2 |V| \rceil$ qubits, and it has only $n + 1$ non-zero amplitudes. The gate count below is for a *level-indexed* layout, which splits the register into a depth counter and a within-level index and therefore costs $\lceil \log_2(n + 1) \rceil + \lceil \log_2 W \rceil$ qubits for maximum level width W — slightly more than the dense layout, by about $\log_2 n$ on $\mathbb{Z}/p^n$. In that layout, preparation is a cascade of n controlled rotations, one per level, each splitting the remaining amplitude between “stop here” and “descend”, followed at each level by writing the within-level index of $\text{anc}_i(x)$ conditioned on the depth register. Both parts together give $O(n \log |V|)$ elementary gates, which is the figure we quote; the n rotations alone are not the whole cost.

Table 1 collects the counts, and the WordNet rows need a word of care that an earlier draft of this paper did not take. The reduced noun hierarchy has 74 374 nodes, but Eq. (14) lives on a tree whose leaves all sit at depth n , and padding WordNet’s shallow leaves out to depth 19

inflates the node count to 673 790 — an order of magnitude, costing 20 qubits rather than the $\lceil \log_2 74\,374 \rceil = 17$ one might expect from the unpadded figure.

Most of that is recoverable. A padded leaf’s chain is unary and private to it, so the $n - d$ basis vectors below its original depth d are shared with no other leaf and contribute nothing to any off-diagonal overlap. Replacing the whole chain by a single node carrying their summed amplitude $\sqrt{1 - \sqrt{f(d)}}$ leaves Eq. (16) untouched and every kernel entry unchanged (test_theorem_c_padding_collapse_preserves_the_kernel). One node per shallow leaf brings WordNet to 132 231 nodes and 18 qubits.

Comparing with Corollary 5.2, on $\mathbb{Z}/p^n$ the construction uses $\log_2 |V| = n \log_2 p + \log_2 \frac{p}{p-1} + o(1)$ qubits while the bound demands at least $n \log_2 p - \log_2 S(p, s)$. For $s > 1$, where $S(p, s)$ is independent of n , the two differ by an additive constant and the encoding is optimal up to $O(1)$ qubits. We resist the stronger phrasing twice over: it is not exactly optimal, and for $s \leq 1$ the lower bound weakens (Remark 5.5) so the gap grows with n .

6.2. The p -adic specialisation

On the regular p -ary tree $\lambda(x, y) = \min(v_p(x - y), n)$, so with $|x - y|_p = p^{-\lambda}$ for $x \neq y$ the geometric profile is $f(\lambda) = p^{-(n-\lambda)s} = (p^{-n}/|x - y|_p)^s$: a decreasing power of the p -adic distance, with $f(n) = 1$ supplying the value at $x = y$. The path state is then a level-weighted superposition over the root-to-leaf path in the p -adic tree, with amplitude concentrated near the leaf for large s and spread towards the root for small s . This is the same object that appears as the state space of continuous-time quantum walks on p -adic trees, where the vertices are sub-balls of a p -adic ball [4, 5], and it connects the encoding to the p -adic quantum formalism of Aniello et al. [6].

6.3. Arbitrary rooted trees

Nothing in the proof uses regularity. The only structural input is Eq. (16), which needs just that ancestors nest, true in any rooted tree. Branching factors may vary between nodes and levels, and shallow leaves are handled by the unary padding of Section 3. The construction therefore applies verbatim to WordNet, whose branching factor ranges from 1 to 402, and $\mathbb{Z}/p^n$ is the homogeneous specialisation rather than the setting (test_theorem_c_generalises_to_a_non_homogeneous_tree).

Numerical verification. Across $(p, n) \in \{(2, 3), (3, 2), (2, 4), (5, 2)\}$ and both geometric and linear profiles, the residual $\max_{x, y} |K - f(\lambda)|$ is at most 2.22×10^{-16} , and the induced distance has zero strong-triangle violations (test_theorem_c_path_states_realise_the_profile_exactly, test_theorem_c_induced_distance_is_exactly_ultrametric).

7. Beyond Exactness: Re-uploading and Noise

7.1. Why an extension must re-upload the data

Theorem 6.1 gives a fixed encoding with no trainable parameters. The obvious way to add expressivity is to follow it with a parametrised circuit, but that does nothing at all: for any unitary $W(\theta)$,

$$|\langle \Phi_f(x) | W(\theta)^\dagger W(\theta) | \Phi_f(y) \rangle|^2 = |\langle \Phi_f(x) | \Phi_f(y) \rangle|^2, \quad (17)$$

so the kernel is invariant. Any non-trivial extension must therefore re-upload the data between layers [7, 8]. We use the family

$$\Phi^{(L)}(x) = W_L D_\gamma(x) \cdots W_1 D_\gamma(x) \Phi_f(x), \quad (18)$$

where $D_\gamma(x)$ applies the phase $e^{i\gamma}$ to every node on the root-to-leaf path of x — the natural data-dependent diagonal on the path-state Hilbert space — and each W_ℓ interpolates between

the identity and a Haar-random unitary by a parameter we call the entangling strength. At $\gamma = 0$ with trivial mixers, Eq. (18) reduces exactly to $\Phi_f(x)$, so the exact construction sits at the origin of the family and anchors any trade-off curve at zero distortion. Note that the innermost $D_\gamma(x)$ phases exactly the support of $\Phi_f(x)$ and so acts as a global phase; the family is non-trivial only because the mixers rotate the state between successive re-uploadings.

We measure two quantities: the *ultrametric distortion*

$$\mathcal{D} = \max_{x,y,z} \max(0, d(x,z) - \max(d(x,y), d(y,z))), \quad d = 1 - K, \quad (19)$$

which vanishes exactly when d satisfies the strong triangle inequality — not quite the same as being an ultrametric, as Remark 7.2 illustrates — and the kernel-target alignment as a standard proxy for usable expressivity. Section 8 maps the resulting Pareto front.

7.2. Depolarising noise preserves the strong triangle inequality

Exactness is a statement about an ideal kernel, so it is fair to ask what survives on hardware. For the dominant global error channel, the geometry survives intact and only the contrast degrades.

Proposition 7.1 (Depolarising noise preserves the strong triangle inequality). *Let K be ultrametric with non-decreasing profile f , realised on a register of Hilbert-space dimension $D_{\mathcal{H}}$ (for the path state in a dense layout, $D_{\mathcal{H}} = 2^{\lceil \log_2 |V| \rceil}$), and let $K_r = (1-r)K + r/D_{\mathcal{H}}$ be the kernel estimated under global depolarising noise of rate $r \in [0, 1]$. Then K_r is level-constant with non-decreasing profile $(1-r)f + r/D_{\mathcal{H}}$, and $d_r = 1 - K_r$ satisfies Eq. (2) for every r .*

Proof. The compute-uncompute estimator reads off the probability of the all-zeros outcome; replacing the state by the maximally mixed state with probability r sends that probability to $1/D_{\mathcal{H}}$, which gives the stated K_r . Hence

$$d_r = 1 - (1-r)K - \frac{r}{D_{\mathcal{H}}} = (1-r)d + r\left(1 - \frac{1}{D_{\mathcal{H}}}\right),$$

an affine function $\varphi(t) = (1-r)t + c$ of d with non-negative slope. A non-decreasing affine map preserves order and commutes with $\max$, so $d(x,z) \leq \max(d(x,y), d(y,z))$ implies $\varphi(d(x,z)) \leq \max(\varphi(d(x,y)), \varphi(d(y,z)))$, which is Eq. (2) for d_r . Applying φ to $K = f(\lambda)$ gives the stated profile, which is still non-decreasing. $\square$

Remark 7.2 (Why this is not quite “ K_r is ultrametric”). For $r > 0$ the profile of K_r has value $(1-r) + r/D_{\mathcal{H}} < 1$ at $v = n$, so it violates the normalisation $f(n) = 1$ of Definition 3.1, and correspondingly $d_r(x, x) > 0$: strictly, d_r is a semi-metric rather than a metric. What the proposition establishes is that the two properties we actually measure — level constancy and the strong triangle inequality — both survive. Rescaling K_r by its diagonal restores $f(n) = 1$ and preserves both, being a positive constant factor. It does not preserve resolution depth, which is thresholded at a fixed τ and so is not scale-free: rescaling inflates every consecutive gap. Resolution figures below are reported on K_r unrescaled.

What depolarising noise destroys is not geometry but contrast. The profile is compressed towards the constant $1/D_{\mathcal{H}}$, so consecutive levels become harder to tell apart and the resolution depth of Definition 3.3 falls, even though level constancy stays exactly zero. Section 8 measures where that collapse happens.

Sampling noise behaves in the opposite way, and the reason is specific to ultrametric spaces. Because every triangle in an ultrametric is isocles with its two longest sides equal, a constant fraction of triples — 0.667 of them on the tree used in our noise experiment — satisfies Eq. (2) *with equality*. An arbitrarily small independent perturbation flips about half of those into violations. Counting violations is therefore an unstable statistic at low noise: it saturates rather than decaying as the shot budget grows. The violation *magnitude* is the meaningful quantity, and it decays as $O(1/\sqrt{\text{shots}})$ as one would expect. We return to this in Section 8.5.

Table 2

The four hierarchies after preprocessing. Source columns describe the full reduced tree; the rest describe the sample the kernels are evaluated on. Qubits is $\lceil \log_2 |V| \rceil$ for the path-state encoding.

dataset	source nodes	source height	sampled leaves	nodes	radix	qubits
synthetic 2-ary	511	8	256	511	2	9
WordNet	74 374	19	256	1 028	11	11
Gene Ontology	10 041	12	256	1 235	16	11
NCBI Mammalia	14 722	14	256	633	38	10

8. Experiments

All experiments are analytic or statevector simulations on CPU, each completing in seconds to minutes. Seeds, resolved configurations and library versions are recorded with every run, and every number below is generated from the experiment outputs rather than transcribed.

8.1. Setup

We use WordNet nouns via NLTK, the molecular-function namespace of the Gene Ontology, the Mammalia clade of the NCBI taxonomy, and synthetic regular p -ary trees for controlled ablations. Table 2 gives the shapes.

Preprocessing, stated rather than assumed. Each source is a directed acyclic graph in the wild, not a tree: WordNet synsets can have several hypernyms. We reduce each to a tree by keeping, for every node, the *longest* path to the root, breaking ties by lexicographically smallest identifier. Multiple inheritance is therefore discarded, which Section 10 records as a limitation. We then truncate every hierarchy at depth 8, because the 2^{height} -dimensional product baselines are otherwise not representable: WordNet reaches depth 19, so an angle or ZZ encoding would need 2^{19} amplitudes per point. Truncation also puts every encoding on the same tree. Finally we sample 256 leaves, restrict to their ancestor closure, and pad shallow leaves to uniform depth. None of these steps changes the lowest-common-ancestor depths among the sampled leaves below the truncation, which is all the kernel depends on.

Baselines. Angle encoding, computational-basis encoding, a ZZ/IQP feature map with two repetitions, a random product map, a block-product map where representable, and two classical references: RBF kernels on integer leaf coordinates and on a one-hot encoding. The geometric profile uses base 2 for the real datasets. Tying the base to the branching factor instead, 38 for NCBI, pushes $f(0)$ to around 10^{-13} , below any tolerance a realistic shot budget could support, so the base is treated as the free design parameter it is.

8.2. E1: ultrametricity and resolution

Table 3 reports both diagnostics of Definition 3.3 on the three real hierarchies; synthetic trees get a systematic sweep in Section 8.4. The path-state encoding attains level constancy 4.44×10^{-16} and zero violations everywhere, at full resolution: the construction is exact on real hierarchies, not only on the regular trees of Theorem 6.1. Every product and entangling baseline has level constancy near 1, angle reaching 0.978 and the random product map 0.991, so their kernels are not imperfect ultrametrics but not ultrametric at all. For the product maps this is Theorem 4.1. For the entangling ZZ map it is consistent with Theorem 5.1 but not implied by it: Corollary 5.2 is stated for the regular p -ary tree, and these hierarchies are irregular, so the ZZ result here is empirical.

Table 3

E1. Level constancy (l.c.) is zero exactly when the kernel is ultrametric with some profile; resolution (res.) counts distinguishable kernel values against the populated levels. Only the path state scores well on both.

encoding	WordNet		Gene Ontology		NCBI Mammalia	
	l.c.	res.	l.c.	res.	l.c.	res.
path state (ours)	4.4×10^{-16}	9/9	4.4×10^{-16}	9/9	4.4×10^{-16}	9/9
basis	0	2/9	0	2/9	0	2/9
angle	0.98	9/9	0.99	9/9	1	9/9
ZZ / IQP	0.48	9/9	0.75	9/9	0.6	9/9
random product	0.99	9/9	0.99	9/9	1	9/9

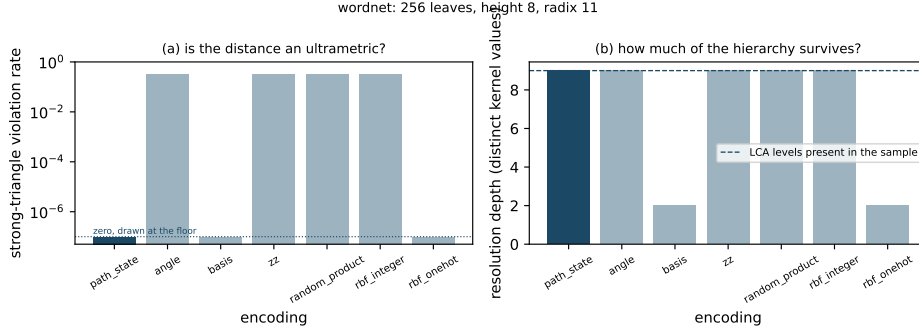


Figure 2: E1 on WordNet. Panel (a) asks whether the induced distance is an ultrametric at all; panel (b) how much of the hierarchy survives. Basis encoding passes (a) perfectly and fails (b) completely, which is why both are reported.

Table 4

E2. Five-fold cross-validated accuracy with a precomputed-kernel SVM at $C = 1$; no hyperparameter was tuned. A dash marks an undefined rank correlation: a delta kernel gives every distinct pair the same distance.

encoding	WordNet		Gene Ontology		NCBI Mammalia	
	acc.	ρ	acc.	ρ	acc.	ρ
path state (ours)	0.79	1	0.6	1	1	1
angle	0.67	0.17	0.56	0.13	0.85	-0.006
ZZ / IQP	0.48	0.098	0.48	0.16	0.9	0.1
random product	0.84	0.37	0.75	0.28	1	0.35
basis	0.21	—	0.28	—	0.68	—
RBF (integer)	0.15	0.0037	0.27	0.027	0.68	0.064

Worth stating plainly, because it complicates the headline: **basis encoding also achieves zero violations**, with level constancy exactly zero and resolution 2 out of 9. This is the $v^* = n$ corner discussed in Section 4, and it is why we do not claim the path state is the unique encoding with an ultrametric kernel. The correct claim is narrower: it is the only one that is both ultrametric and fully resolving.

8.3. E2: hierarchical classification

Table 4 predicts a leaf’s depth-2 ancestor and correlates the kernel’s induced distance with true tree distance. Classes with fewer than five members are dropped, with the count recorded in the run metadata.

The path state attains $\rho = 1$ on every dataset. This is a consequence of Eq. (15), not a fitted result: the induced distance is a strictly monotone function of tree distance, so perfect rank correlation follows. No baseline exceeds 0.368.

On accuracy the picture is different, and we report it as it came out. On WordNet the path

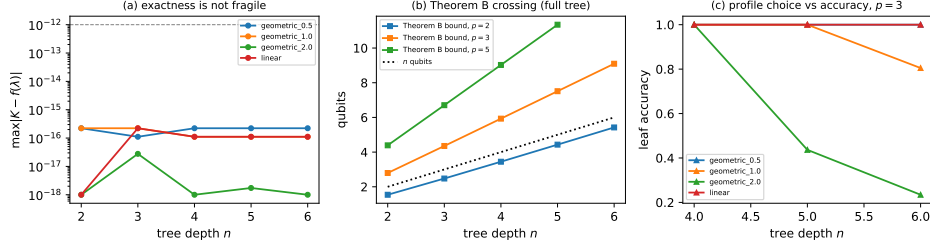


Figure 3: E3. (a) The profile residual stays at machine precision for every strictly monotone profile at every depth. (b) The Theorem 5.1 bound against the n qubits a product or IQP encoding provides: above the diagonal for $p \geq 3$, below it for $p = 2$. (c) Profile choice against accuracy.

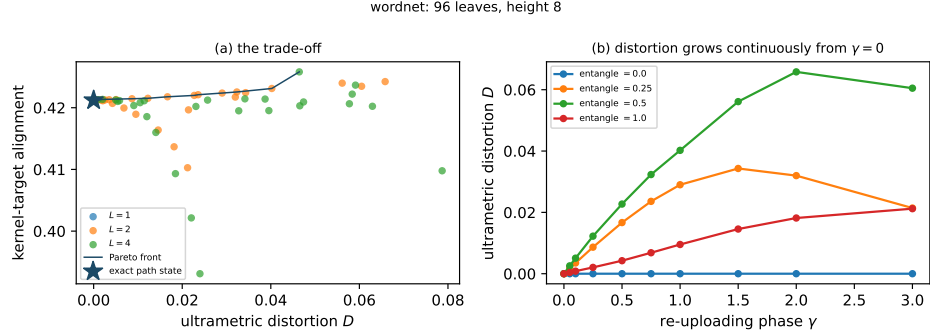


Figure 4: E4 on WordNet, over 120 configurations of Eq. (18). Distortion grows continuously from zero, so the exact construction is the endpoint of a tunable family rather than an isolated point; the alignment bought in exchange is small.

state reaches 0.792 ± 0.0157 while the random product baseline reaches 0.839, with a similar ordering on Gene Ontology. A kernel that faithfully encodes tree distance is therefore not the best kernel for this task, which is unsurprising once stated: the label is the depth-2 ancestor, and a kernel spending capacity on depths 3 through 8 spends it on distinctions the label ignores. The comparison against v-PuNNs [9], at 99.96% leaf accuracy on WordNet, is not close and must be read the same way: that is a trained deep model with a bespoke optimiser over p -adic weight space, this a fixed encoding with a closed form and no parameters. Geometric fidelity and task accuracy are different objectives; this paper claims the former. The criterion was fixed before any number was produced, so that a weak result could not tempt a retrofit of the claims.

8.4. E3 and E4: ablations and the re-uploading trade-off

Figure 3 sweeps profile shape, depth and branching factor over 56 configurations. Panel (a) is the robustness check for Theorem 6.1: the worst residual is 2.22×10^{-16} and the total violation count is 0, so exactness degrades with neither depth, radix nor profile shape. Panel (b) is Corollary 5.2 in picture form. For $p = 3$ and $p = 5$ the required qubit count runs above the diagonal at every depth; for $p = 2$ it runs below, which is Remark 5.6. The curves use the closed form Eq. (9) on the full tree, since a subsampled estimate saturates at $\log_2$ of the sample size.

E4 answers the structural question about Eq. (18) affirmatively: distortion grows continuously from zero, so the exact construction is the zero-distortion endpoint of a continuum rather than an isolated case. The quantitative answer is less encouraging. The exact point achieves alignment 0.421, and the best configuration anywhere in the family achieves 0.426 at distortion 0.0464, a gain of 0.00459. On these datasets the trade-off exists but buys little, and we would not recommend paying for it.

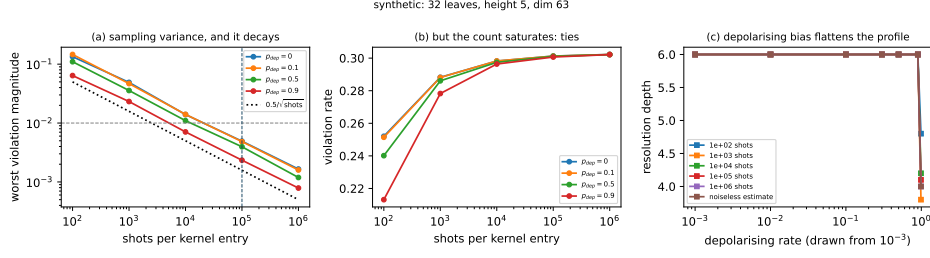


Figure 5: E5. (a) The violation magnitude decays as $O(1/\sqrt{\text{shots}})$, tracking the dotted reference. (b) The violation *count* saturates instead, because a constant fraction of triples in an ultrametric is exactly tight. (c) Depolarising noise leaves the geometry intact but flattens the profile until resolution collapses.

8.5. E5: noise and finite shots

Proposition 7.1 predicts that depolarising noise cannot break the strong triangle inequality, and the measurement agrees exactly: at every rate up to 0.99 the noiseless-estimate row has zero violations and zero excess. Resolution degrades, and only at extreme rates.

Finite sampling is the real constraint, and measuring it correctly required abandoning the obvious statistic. The violation count *increases* with the shot budget towards a plateau, which looks like a bug and is not: 0.667 of triples here meet Eq. (2) with equality, and any perturbation flips about half of them, so the count converges to roughly half the tie fraction rather than to zero — the observed plateau near 0.30. The magnitude behaves as expected, falling from 0.134 at 100 shots to 0.00165 at 1 000 000, a factor of $\sqrt{10}$ per decade. Defining the budget on the magnitude gives 100 000 **shots per kernel entry** for a worst-case excess below 0.01, over 10 repetitions.

9. Related Work

p -adic quantum mechanics. Quantum systems over p -adic fields have a developed formalism. Aniello et al. [6] construct a model of quantum states over a quadratic extension of $\mathbb{Q}_p$, working the p -adic qubit and quNit out explicitly. Zúñiga-Galindo [4] and Zúñiga-Galindo [5] study continuous-time quantum walks whose vertex set consists of the sub-balls of a p -adic ball, that is, on the nodes of the p -adic tree, and Zúñiga-Galindo and Chacón-Cortés [10] relate discretised p -adic Schrödinger dynamics to continuous-time quantum Markov chains. Our path states live in $\mathbb{C}^V$ with V the node set of that same tree, so the encoding of Theorem 6.1 is a state in the Hilbert space these works take as their setting. We use it statically, as a feature map, rather than evolving it; connecting the profile f to the spectrum of a p -adic walk generator is a natural next question that we do not address.

p -adic machine learning. The closest prior work is v-PuNNs [9], which builds neurons as characteristic functions of p -adic balls, proves a finite hierarchical approximation theorem, and reports strong results on WordNet, Gene Ontology and the NCBI taxonomy. Our benchmarks are taken from that work so the comparison is direct. The relationship is complementary rather than competitive: v-PuNNs is a trained architecture over p -adic weight space, while this paper characterises which *quantum encodings* can carry ultrametric structure at all. v-PuNNs is accordingly far more accurate on the shared benchmarks, as we report in Section 8.3. Separately, Bradley [3] asks how ultrametric real finite metric data actually is, recasting Vietoris–Rips complexes via p -adic Mumford curves; that question is prior to ours, which assumes the target geometry is ultrametric and asks what can represent it.

Quantum kernels. The fidelity kernel construction is due to Havlíček et al. [2], and Schuld [1] showed that supervised quantum models are kernel methods, which is what makes the encoding the locus of inductive bias. A recent line of work establishes limits on what these kernels can do: exponential concentration as system size grows [11], double descent [12] and benign overfitting [13] in their generalisation behaviour, and trainable embeddings as a response [8]. Those results concern how a quantum kernel behaves once chosen. Theorems 4.1 and 5.1 concern which kernels are reachable in the first place, and are in that sense upstream: our obstructions hold before any question of trainability or concentration arises. Theorem 5.1 in particular does not care whether the encoding concentrates, only how wide it is.

Hierarchy in classical representation learning. Hyperbolic embeddings are the established answer to representing trees [14], and the approach has grown into a substantial literature [15]. Hyperbolic space achieves low but non-zero distortion of tree metrics, because a tree embeds into it only approximately. The path-state construction achieves zero distortion by design, at the cost of a dimension linear rather than logarithmic in the number of nodes. By Theorem 5.1, that cost is not an artefact of our construction but a lower bound for any exact realisation.

Data re-uploading. Interleaving data encodings with trainable layers is due to Pérez-Salinas et al. [7] and remains actively developed [16]. We use it in Section 7 not to increase expressivity for its own sake but as the minimal way to extend a fixed encoding at all, given the invariance Eq. (17).

10. Limitations and Outlook

Scope of the claims. Restating the point of Section 1 where a reader will look for it: $K(x, y) = f(\lambda(x, y))$ is computable classically in time linear in the tree depth, so nothing here accelerates kernel evaluation. The obstructions constrain quantum encoding design and the construction is useful as an input stage for deeper models; neither is a speedup claim. Two further boundaries are worth naming. Theorem 5.1 is strong when the profile decays quickly and weak when it does not: Remark 5.5 shows $s > 1$ buys a uniform threshold rather than the exclusion itself, and gives the exact criterion Eq. (12): below $\min(s, 1) = \log_p 2$, and at $p = 2$ for every s , the bound stops forbidding narrow encodings, and we do not know whether they exist there. And as Section 8.3 reports, the path state loses on classification accuracy to both a random product map and to v-PuNNs; the paper’s claims rest on Theorems 4.1, 5.1 and 6.1 and on E1, not on E2.

Preprocessing discards real structure. All three hierarchies are directed acyclic graphs, and the longest-path reduction of Section 8.1 throws away every hypernym but one, so a synset with two genuinely distinct parents is misrepresented. Extending the construction to DAGs is not a matter of choosing a better reduction: ancestor sets in a DAG need not be nested, and Eq. (16) — the entire content of Theorem 6.1 — depends on that nesting. Separately, comparing against 2^{height} -dimensional baselines forced a depth-8 truncation. The path-state encoding has no such limit, Table 1 giving 18 qubits for untruncated WordNet, so this constrains the comparison rather than the method; we have not measured the baselines at full depth because they cannot be represented there.

Simulation only. Everything here is statevector simulation. Proposition 7.1 covers the global depolarising channel analytically and Section 8.5 gives a shot budget of 100 000 per kernel entry, but coherent errors, readout error and device-specific noise are untested. The budget is per entry, so a full $m \times m$ kernel matrix costs $O(m^2)$ times that, which is the practical barrier to hardware use at scale.

Outlook. The most immediate direction is using the path state as the input layer of a deeper variational model, where Section 8.4 suggests the interesting regime is small distortion rather than large. The second is the p -adic connection: the encoding lives in the state space of quantum walks on the p -adic tree [4], and relating the profile f to the spectrum of a walk generator would give a dynamical reading of what is here a static construction.

11. Conclusion

The geometry a quantum kernel imposes on its data is decided by the encoding, before training begins. For hierarchical data that geometry should be ultrametric, and we have shown that the encodings in common use cannot make it so. Product feature maps fail for a structural reason — multiplicativity across digits is incompatible with resolving a tree beyond a bounded depth — and encodings of insufficient width fail for a reason that does not depend on their structure at all, which is what rules out the entangling maps the first argument cannot reach.

The positive result is that path states close the gap exactly. They realise precisely the strictly monotone ultrametric kernels, on arbitrary rooted trees, using a number of qubits that, on $\mathbb{Z}/p^n$ with a rapidly decaying profile, is within an additive constant of the proven lower bound, with $n + 1$ non-zero amplitudes and $O(n \log |V|)$ gates. Global depolarising noise acts on the induced distance as a non-decreasing affine map and so preserves level constancy and the strong triangle inequality at every rate, though not the normalisation that makes the noisy kernel ultrametric in our sense. Finite sampling does break the inequality, and the violation magnitude falls below one per cent of the kernel range at a measured 100 000 shots per kernel entry.

What this does not deliver is a faster kernel. The construction’s value is as a constraint on encoding design and as an input stage for models where no closed form survives, and as an argument that the encoding deserves to be chosen for the geometry of the data rather than for the convenience of the circuit.

Declaration on Generative AI

Complete per the CEUR-WS generative-AI disclosure policy in force at submission.

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